Determining g

AS Level | **9702 AS Ch.2** | Chapter 2: Accelerated motion | Lesson 2.12 | Sections 2.12

Goal: Describe a free-fall experiment to find g, and get g from a graph of s against t².

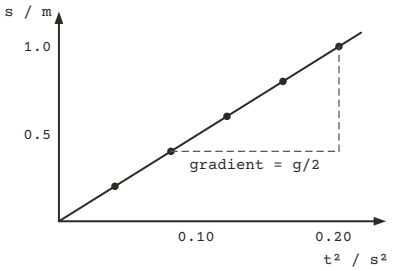
01 The big picture

Drop a small steel ball from rest from an electromagnet. Switching off the magnet starts a timer; the ball hitting a trapdoor stops it. Measure the height s with a metre rule and repeat each height, averaging t.

From rest, $s = \frac{1}{2}gt^2$, so a graph of s against t^2 is a straight line through the origin with gradient $g/2$. Light gates with a falling card, or video, can also be used.

Free fall [paraphrase]
Falling with only the weight acting: acceleration g, no air resistance.

Fig. 1 · s against t²



A straight line through the origin: gradient $= g/2$.

02 Key equations

From rest
$$s = \tfrac{1}{2}gt^2$$

One drop
$$g = \frac{2s}{t^2}$$

From the graph
$$g = 2 \times \text{gradient of } s\text{–}t^2$$

Symbols and units

s	height fallen	m	t	time to fall	s	g	acceleration of free fall	$\mathrm{m\,s^{-2}}$

03 Sample problems with solutions

Sample 1 · One drop

A ball falls 1.20 m from rest in a mean time of 0.495 s. Calculate g.

Step 1. Rearrange $s = \frac{1}{2}gt^2$:

$$g = \frac{2s}{t^2} = \frac{2 \times 1.20}{0.495^2} = 9.79\ \mathrm{m\,s^{-2}}$$

Answer: $g = 9.79\ \mathrm{m\,s^{-2}}$ (3 s.f.)

Unit check: $\frac{\mathrm{m}}{\mathrm{s^2}} = \mathrm{m\,s^{-2}}$ ✓

04 Practice problems

P1 The gradient of a graph of s against t^2 is 4.95 m s⁻². Find g.
Answer: 9.90 m s⁻²

P2 A ball falls 0.80 m from rest in 0.40 s. Find g.
Answer: 10 m s⁻²

P3 Why use a small, dense ball?
Answer: Air resistance is negligible compared with its weight

P4 Why repeat each height and average the times?
Answer: To reduce the effect of random errors

05 Other useful information

Common mistakes

✗ **Wrong:** Plot s against t and take the gradient as g.
✓ **Right:** That curves; plot s against t^2 and double the gradient.

Exam tips

- Measure s from the bottom of the ball to the trapdoor.

What the syllabus asks

- 9702 topic 2: describe an experiment to find g using a falling object. Coursebook 2.12.

Explore: Practice questions for this lesson